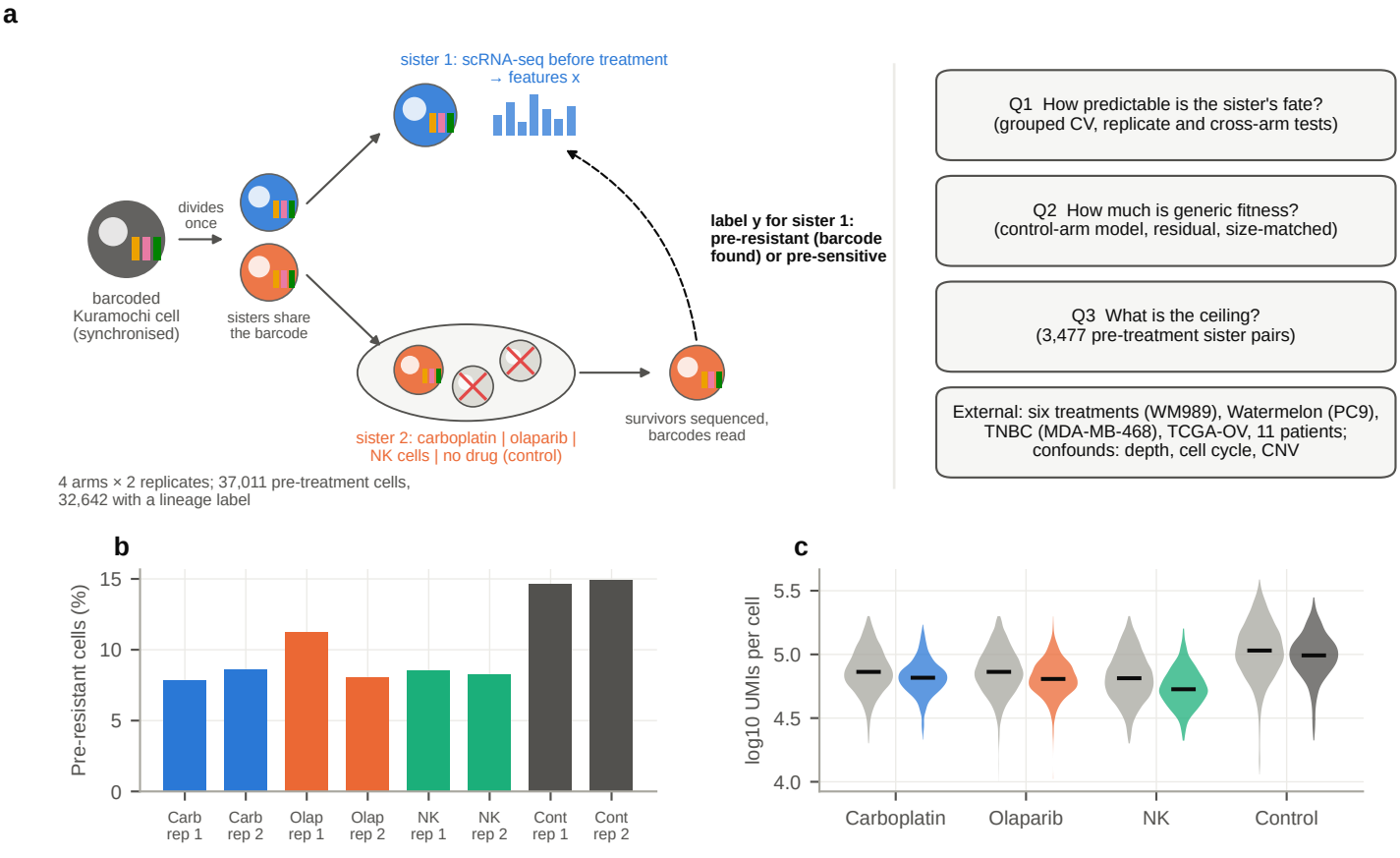
a
sister 1: scRNA-seq before treatment
→ features x
divides
once
barcoded
Kuramochi cell
(synchronised)
sisters share
the barcode
label y for sister 1:
pre-resistant (barcode
found) or pre-sensitive
sister 2: carboplatin | olaparib |
NK cells | no drug (control)
survivors sequenced,
barcodes read
4 arms × 2 replicates; 37,011 pre-treatment cells,
32,642 with a lineage label
Q1 How predictable is the sister's fate?
(grouped CV, replicate and cross-arm tests)
Q2 How much is generic fitness?
(control-arm model, residual, size-matched)
Q3 What is the ceiling?
(3,477 pre-treatment sister pairs)
External: six treatments (WM989), Watermelon (PC9),
TNBC (MDA-MB-468), TCGA-OV, 11 patients;
confounds: depth, cell cycle, CNV
b
Pre-resistant cells (%)
15
10
5
0
Carb rep 1
Carb rep 2
Olap rep 1
Olap rep 2
NK rep 1
NK rep 2
Cont rep 1
Cont rep 2
c
log10 UMIs per cell
5.5
5.0
4.5
4.0
Carboplatin
Olaparib
NK
Control